# Ollivier–Ricci Curvature as a Landscape Analysis Framework for Discrete Combinatorial Optimization

Roberto Sergiu Hordoan[0000−0000−0000−0000]

Babeş–Bolyai University, Cluj-Napoca, Romania
roberto.hordoan@stud.ubbcluj.ro

**Abstract.** We introduce the *ORC Transition Graph* (OTG), a deterministic directed graph over local optima derived from Ollivier–Ricci curvature (ORC), as a structural analysis tool for discrete fitness landscapes. Each local optimum has exactly one outgoing edge to the optimum reached by following its most-negative-curvature direction and hill-climbing—a purely local, $O(k^2)$ computation. We study OTG on NK landscapes, epistatic W-model benchmarks ($N = 16$, full enumeration), random MAX-SAT ($N$ up to 100), QAP (permutation swaps, $n$ up to 9), and TSP (2-opt, $n$ up to 50 via sampling). The OTG extracts cross-basin information invisible to standard local optima networks (LONs): using only 1-hop neighborhood data, it achieves up to 9.2× better attractor quality than LON-d1 at equal information cost on epistatic W-model landscapes, and even outperforms the stochastic 3-hop LON-d3 on those instances. A structural crossover in OTG topology—from acyclic to cyclic at moderate epistasis—correlates with when ORC-guided perturbation outperforms random perturbation in iterated local search (ILS). The OTG framework applies to any graph with the disjoint-neighborhood property (Proposition 1), which we prove covers binary hypercubes, permutation swap graphs, *and* TSP 2-opt neighborhoods—the three foundational neighborhoods in combinatorial optimization. ORC's per-step escape advantage grows monotonically with neighborhood size—from 3.1× to 18.3× over random on TSP ($n = 8$ to 50)—though its computational cost ($O(k^2)$ per optimum) limits algorithmic use to settings where fitness evaluations are expensive. Per-optimum ORC metrics converge from ∼20 sampled optima (<1% error), while full OTG structural analysis remains limited to enumerable landscapes.

**Keywords:** Fitness landscape analysis · Ollivier–Ricci curvature · Local optima networks · Discrete optimization · Permutation spaces

## 1 Introduction

Fitness landscape analysis (FLA) provides the theoretical foundation for understanding why metaheuristic optimizers succeed or fail on specific problem

instances [16] and for automated algorithm selection [4]. Classical metrics—fitness-distance correlation (FDC) [2], autocorrelation length [19], information content [23], and exploratory landscape analysis (ELA) features [8]—characterize global statistical properties such as ruggedness and smoothness. Local optima networks (LONs) [11,13] go further by representing basin connectivity as a graph whose edges are discovered through stochastic perturbation followed by hill climbing (HC).

Both approaches share a limitation: classical metrics provide scalar summaries blind to local geometric structure, while LONs provide connectivity but not *directional* information—at a local optimum, which specific neighbor escapes the basin? LON construction also depends on perturbation distance $d$ and requires many stochastic trials per optimum.

We address these limitations through Ollivier–Ricci curvature (ORC) [15], a discrete analogue of Ricci curvature that measures how neighborhood distributions of adjacent nodes diverge or converge. Negative curvature identifies directions where the landscape geometry "spreads apart"—the signature of inter-basin saddle regions. From ORC, we construct the *ORC Transition Graph* (OTG): a deterministic directed graph where each local optimum points to the optimum reached by following its min-ORC direction.

**Contributions.**

1. We formalize fitness-lifted ORC for discrete optimization landscapes and prove that negative curvature is guaranteed at every local optimum on graphs with the disjoint-neighborhood property, covering binary hypercubes, permutation swap graphs, and TSP 2-opt neighborhoods.
2. We introduce the OTG, a deterministic, parameter-free directed graph over local optima derived from optimal-transport-based curvature. On epistatic binary landscapes, the OTG achieves up to 9.2× better attractor quality than LON-d1 at equal information cost; across binary, permutation, and tour search spaces, ORC's per-step escape advantage grows monotonically with neighborhood size.
3. We validate the framework algorithmically through ORC-guided iterated local search (ILS) and show that OTG structural properties—specifically the presence or absence of cycles—indicate when ORC-guided perturbation adds value over random perturbation.

## 2 Background

A *fitness landscape* $(S, \mathcal{N}, f)$ consists of solutions $S$, neighborhood function $\mathcal{N} : S \to 2^S$, and fitness $f : S \to \mathbb{R}$ [19]. A solution $x^*$ is a *local optimum* (minimization) if $f(x^*) \leq f(y)$ for all $y \in \mathcal{N}(x^*)$. LONs [11,12] compress the landscape into a graph whose nodes are local optima and whose edges encode transitions discovered by perturbing a solution (flipping $d$ random bits) and hill-climbing. LON features—including funnel structure [14], fractal dimension [20],

and phase-transition indicators [10]—predict search difficulty [13], but edge construction requires $T$ stochastic trials per optimum (typically $T \geq 50$) and the resulting funnel decomposition depends on $d$.

Ollivier [15] defined Ricci curvature for metric-measure spaces; Lin et al. [6] gave a modified graph formulation. For a graph edge $(x, y)$:

$$\kappa(x, y) = 1 - \frac{W_1(\mu_x, \mu_y)}{d(x, y)} \tag{1}$$

where $\mu_x, \mu_y$ are neighborhood probability measures and $W_1$ is the Wasserstein-1 distance [24]. Negative curvature signals diverging neighborhoods—a signature of community bridges [9] and GNN bottlenecks [22,18]; positive curvature signals contraction. We choose Ollivier's formulation over Forman's [17]: on triangle-free graphs (including all three neighborhoods in Corollary 1), Forman curvature is the constant $4-2k$ for every edge on unweighted $k$-regular graphs, and depends only on endpoint-local quantities even with fitness weighting—missing the distributional matching that gives ORC its cross-basin signal. To our knowledge, ORC has not been applied to fitness landscape analysis.

## 3 ORC on Fitness Landscapes

### 3.1 Fitness-Lifted ORC

We define ORC on the search graph $(S, \mathcal{N})$ with uniform measures $\mu_x$ over $\{x\} \cup \mathcal{N}(x)$. The transport cost between $a \in \text{supp}(\mu_x)$ and $b \in \text{supp}(\mu_y)$ is:

$$c(a, b) = \delta(a, b) + \gamma \cdot |f(a) - f(b)| \tag{2}$$

where $\gamma \geq 0$ controls fitness sensitivity and the overlap indicator is:

$$\delta(a, b) = \begin{cases} 0 & \text{if } a = b \\ 1 & \text{if } a \text{ or } b \in \text{supp}(\mu_x) \cap \text{supp}(\mu_y) \\ 2 & \text{otherwise (both exclusive)} \end{cases} \tag{3}$$

This is a conservative coarse-graining of graph distance computable from 2-hop information alone, preserving $O(k^2)$ locality. For uniform measures of equal support size, $W_1$ reduces to an optimal assignment problem solved by the Hungarian algorithm in $O(k^3)$ [5]; the fine-grained complexity of this reduction has been formally established [1].

**Parameter sensitivity.** We use $\gamma = 1$ throughout. Importantly, the ORC-guided *direction* (Algorithm 1) is $\gamma$-invariant: changing $\gamma$ scales all fitness differences uniformly, so the optimal transport matching order—and thus the min-ORC neighbor—does not change. Empirically, escape rates are identical to three decimal places across $\gamma \in \{0.1, 0.5, 1.0, 2.0\}$ on all 480 instances; only absolute curvature values change. The framework is thus parameter-free for navigation; $\gamma$ matters only for difficulty prediction, where $\gamma = 1$ balances structural and fitness terms. This invariance is a consequence of the *linear* fitness term in Eq. (2): a nonlinear cost $g(|f(a) - f(b)|)$ would break it, though such extensions are a natural direction for future work.

### 3.2 Why ORC Sees What 1-Hop Methods Cannot

**Proposition 1 (ORC Cross-Neighborhood Decomposition).** *Let $G = (V, E)$ be a regular graph of degree $k$ in which adjacent nodes share no common neighbors: $\mathcal{N}(x) \cap \mathcal{N}(y) = \emptyset$ for all $(x, y) \in E$. Then $|\mathrm{supp}(\mu_{x^*}) \cap \mathrm{supp}(\mu_y)| = 2$ (only $x^*$ and $y$ are shared), and the optimal transport cost decomposes as:*

$$(k{+}1)\, W_1(\mu_{x^*}, \mu_y) = C_{\text{shared}} + \min_{\sigma} \sum_{i=1}^{k-1} \big[2 + \gamma\, |f(a_i) - f(b_{\sigma(i)})|\big] \tag{4}$$

*where $C_{\text{shared}} \leq 2 + \gamma(\Delta + \Delta')$ accounts for the two shared elements, $\{a_i\} = \mathcal{N}(x^*) \setminus \{y\}$, $\{b_j\} = \mathcal{N}(y) \setminus \{x^*\}$ are the $k{-}1$ exclusive neighbors on each side, and $\sigma$ ranges over permutations of $\{1, \ldots, k{-}1\}$.*

*Proof.* Since $\mathcal{N}(x^*) \cap \mathcal{N}(y) = \emptyset$, the only shared support elements are $x^*$ and $y$ themselves. Any assignment $\pi : \mathrm{supp}(\mu_{x^*}) \to \mathrm{supp}(\mu_y)$ partitions into: (i) the two shared positions $(x^*, y)$ contributing $C_{\text{shared}}$, and (ii) the $k$–1 exclusive pairs, each incurring $\delta = 2$ plus a fitness-difference term. The optimal $\sigma$ minimizes the total fitness-comparison cost.

**Corollary 1 (Applicability to Standard Search Spaces).** *Proposition 1 applies to: **(i)** the $N$-dimensional binary hypercube with bit-flip neighborhood ($k = N$), **(ii)** the symmetric group $S_n$ with pairwise-swap neighborhood ($k = \binom{n}{2}$), and **(iii)** Hamiltonian tours with 2-opt neighborhood ($k = n(n{-}3)/2$).*

*Proof.* **(i)** On the hypercube, flipping bit $i$ in $x$ and bit $j$ in $y = x \oplus e_p$ yields the same neighbor only if $e_i = e_p \oplus e_j$; since single bit-flips cannot equal a two-bit change, $\mathcal{N}(x) \cap \mathcal{N}(y) = \emptyset$. **(ii)** On $S_n$, if $y = x \circ \tau_{ab}$, a common neighbor $z$ requires $\tau_{ij} = \tau_{ab} \circ \tau_{kl}$ for transpositions $\tau_{ij}$, $\tau_{kl}$. The product of two distinct transpositions is a 3-cycle or double transposition—never a transposition. **(iii)** A 2-opt move is a segment reversal $\rho_{p,q}$ mapping position $i \mapsto p{+}q{+}1{-}i$ for $i \in \{p{+}1, \ldots, q\}$ and fixing all others—an affine map with *slope* $-1$ on its support. A common neighbor requires $\rho_{a,b} = \rho_{c,d} \circ \rho_{p,q}$ for distinct non-trivial reversals. *Disjoint supports:* the composition has non-contiguous support (two separated intervals with a fixed gap), impossible for a single reversal. *One nested in the other* (WLOG $\mathrm{supp}(\rho_{p,q}) \subset \mathrm{supp}(\rho_{c,d})$): on the inner interval, $(\rho_{c,d} \circ \rho_{p,q})(i) = i + \delta$ is a *translation* (slope $+1$), since each reversal contributes slope $-1$ and $(-1)(-1) = +1$. Any single reversal has slope $-1$ on its entire support, so the composition cannot be one. *Partial overlap:* within the overlap the composition transitions from slope $+1$ (where both act) to slope $-1$ (where only one acts)—again incompatible with a single reversal. Verified computationally at $n \leq 8$ (all 25,200 edges) and by sampling at $n \leq 11$ (all >200,000 tested edges have overlap 0).

These three search spaces—binary strings (SAT, NK), permutations with swaps (QAP, scheduling), and tours with 2-opt (TSP)—cover the foundational neighborhoods in combinatorial optimization.

**Algorithm 1** ORC-Guided Escape from a Local Optimum

**Require:** Local optimum $x^*$, fitness $f$, neighborhood $\mathcal{N}$, parameter $\gamma$
1: **for** each $y \in \mathcal{N}(x^*)$ **do**
2: Compute $\kappa(x^*, y)$ via Eq. (1)–(3)
3: **end for**
4: $y^* \leftarrow \arg\min_y \kappa(x^*, y)$
5: **return** HillClimb($y^*$)

The exclusive-pair term is the key: it computes an optimal matching between the fitness distributions visible from $x^*$ and from $y$. At a basin boundary, the two sides' fitness distributions diverge, making curvature strongly negative. By contrast, LON-d1 evaluates each neighbor $y$ using only $f(y)$ and the downstream hill-climbing destination, discarding the $(k-1)$ exclusive neighbors entirely—precisely the distributional comparison that ORC captures.

### 3.3 Negative Curvature Guarantee

**Theorem 1.** *Let $x^*$ be a strict local optimum on any degree-$k$ graph satisfying the disjoint-neighborhood condition of Proposition 1 ($k \geq 4$), with cost* (2)–(3) *and $\gamma > 0$. If neighbor $y$ satisfies $\Delta = f(y) - f(x^*) < (k-3)/((k+1)\gamma)$, then $\kappa(x^*, y) < 0$.*

*Proof.* From Proposition 1, every assignment incurs exclusive-pair structural cost at least $2(k-1)$, giving $W_1 \geq 2(k-1)/(k+1)$. With edge cost $c(x^*, y) = 1 + \gamma\Delta$: $\kappa(x^*, y) \leq 1 - 2(k-1)/[(k+1)(1+\gamma\Delta)]$, which is negative iff $\Delta < (k-3)/((k+1)\gamma)$.

For the binary hypercube with $k = N = 16$, $\gamma = 1$: the bound requires $\Delta < 0.76$. For QAP with $n = 8$ ($k = 28$): $\Delta < 0.86$. Across all 5,000+ local optima in our binary benchmarks and 1,500+ QAP optima, **100% satisfy the condition**. The min-ORC neighbor $y^* = \arg\min_y \kappa(x^*, y)$ identifies the direction of maximal neighborhood divergence—the strongest saddle direction (Alg. 1).

### 3.4 ORC-Guided Escape and the OTG

Applying Algorithm 1 at every local optimum of a landscape defines the *ORC Transition Graph* (OTG): a deterministic directed graph where each optimum has exactly one outgoing edge to the optimum reached by following its min-ORC neighbor and hill-climbing. The OTG is a functional graph; its terminal attractors are either self-loops (sinks) or directed cycles. The *funnel* of an attractor is the set of all optima whose OTG path converges to it. Unlike LONs [21], the OTG is deterministic and parameter-free: it requires no perturbation distance or stochastic trials.

# 4 Experimental Setup

## 4.1 Benchmark Problems

We evaluate on five benchmark families spanning three search spaces:

**NK landscapes** [3] define tunably rugged fitness via $K$ epistatic interactions per bit on length-$N$ binary strings. We test $K \in \{4, 8, 12\}$ at $N = 16$ (30 instances each, full enumeration).

**W-model** [25] applies neutrality, epistasis, and ruggedness transformations controlled by a difficulty parameter $\nu$ to a base OneMax problem. We test $\nu \in \{3, 4, 6\}$ at $N = 16$ (30 instances each, full enumeration).

**Random MAX-SAT.** Random 3-SAT instances with $N \in \{20, 50, 100\}$ variables and clause-to-variable ratio $\alpha \in \{3.0, \ldots, 6.0\}$ (20 instances each). Local optima are collected via random-restart HC (500–2,000 restarts). This domain has well-studied phase transitions [10] and no structural overlap with NK or W-model.

**Random QAP.** Quadratic assignment problems with uniform random flow/distance matrices and pairwise-swap neighborhood, $n \in \{6, 7, 8, 9\}$ (20 instances each, full enumeration).

**Random TSP.** Random-matrix TSP instances with 2-opt neighborhood. Full enumeration at $n \in \{8, 9, 10\}$ (20 each, 4–7 optima per instance); sampling-based analysis at $n \in \{15, 20, 30, 50\}$ (20 each, 300 random-restart 2-opt runs per instance). TSP instances at small $n$ serve to validate the theoretical framework on an exact ground truth; sampling-based runs at larger $n$ test scaling.

The five families were chosen to cover three distinct search spaces (binary strings, permutations, tours), a range of landscape ruggedness levels, and both enumerable and sampling-only regimes.

## 4.2 Algorithms and Baselines

**Hill Climbing (HC).** Best-improvement (steepest descent) HC: at each step, all $k$ neighbors are evaluated and the one with the best fitness is selected. HC terminates when no neighbor improves on the current solution, yielding a local optimum. Ties are broken uniformly at random.

**Random-Restart Hill Climbing (RR-HC).** Repeat HC from uniformly random initial solutions until the budget of fitness evaluations (FEs) is exhausted. Return the best local optimum found.

**Iterated Local Search (ILS)** [7]. From a local optimum, apply a perturbation (bit-flips) and re-run HC; repeat until the FE budget is exhausted. We test four ILS variants, all using a fixed perturbation strength of 3 bit-flips for fair comparison:

- **Rand-ILS**: flip 3 random bits.
- **ORC+Pert**: flip the min-ORC bit (Algorithm 1) plus 2 random bits.
- **ORC-ILS**: flip only the min-ORC bit (1 bit; no random component).
- **MG-ILS**: flip only the MinGap bit (1 bit; see below).

**Table 1.** ORC escape direction analysis on binary landscapes ($N = 16$, full enumeration). %ORC: fraction of local optima where Algorithm 1 reaches a better optimum. %MinGap: same using the argmin-$|\Delta|$ heuristic. %Rand: random neighbor baseline (30 trials). The ORC advantage over MinGap emerges on epistatic W-model landscapes where the fitness-distance relationship is distorted.

| Config | #Opt | %ORC | %MinGap | %Rand | ORC/R | ORC/MG | Agree |
|---|---|---|---|---|---|---|---|
| NK $K=4$ | 170 | 59.6 | 57.4 | 31.6 | 1.9× | 1.04× | 89.6% |
| NK $K=8$ | 888 | 58.5 | 56.5 | 33.6 | 1.7× | 1.04× | 90.3% |
| NK $K=12$ | 2249 | 51.1 | 49.4 | 37.6 | 1.4× | 1.03× | 91.7% |
| W $\nu=3$ | 406 | 79.9 | 43.5 | 29.4 | **2.7**× | **1.84**× | 59.6% |
| W $\nu=4$ | 3732 | 59.7 | 25.9 | 39.5 | 1.5× | **2.31**× | 43.3% |
| W $\nu=6$ | 5836 | 63.9 | 37.4 | 48.6 | 1.3× | 1.71× | 52.7% |

ORC-ILS and MG-ILS use only 1-bit perturbation; they are included as ablations to isolate the effect of the directed bit.

**MinGap.** The neighbor $y$ with the smallest fitness difference $|f(y) - f(x^*)|$ (argmin-$|\Delta|$ heuristic). This greedy baseline uses only fitness values, not distributional information.

**LON-d1.** At a local optimum $x^*$, the LON-d1 destination is the mode HC outcome over all $k$ single-bit-flip neighbors (1-hop, deterministic).

**LON-d3.** At a local optimum $x^*$, the LON-d3 destination is the mode HC outcome from 100 random 3-bit perturbations (3-hop, stochastic).

### 4.3 Evaluation Metrics

**Escape rate (%).** Fraction of local optima where a method reaches a strictly better optimum after one perturbation + HC step. This is the primary metric shared across all experiments.

**Mean terminal rank.** Normalized fitness rank of the attractor reached by following OTG/LON paths (0.0 = best, 0.5 = random).

**Top-5% attractor (%).** Fraction of optima whose OTG/LON path terminates at an attractor in the top 5% by fitness.

**Compression.** #funnels / #optima—lower means better funnel aggregation.

**Success rate (%).** Fraction of ILS runs finding the global optimum within the FE budget (used for the ILS comparison in Section 5.4).

## 5 Results

### 5.1 Single-Step Escape Analysis

Table 1 presents the single-step escape analysis. On NK landscapes, ORC beats random by 1.4–1.9× but only marginally outperforms MinGap (1.03–1.04×); on

**Table 2.** OTG vs. LON funnel quality ($N = 16$, full enumeration). LON-d1: 1-hop, deterministic. LON-d3: 3-hop, 100 stochastic trials. Rank: 0.0 = best, 0.5 = random. Compression: #funnels / #optima.

| | | Mean terminal rank | | | Top-5% attractor | | | Compression | | |
|---|---|---|---|---|---|---|---|---|---|---|
| **Config** | **#Opt** | **OTG** | **d1** | **d3** | **OTG** | **d1** | **d3** | **OTG** | **d1** | **d3** |
| NK $K=4$ | 170 | .255 | .490 | **.181** | 24.6 | 5.6 | **30.1** | 40 | 98 | 35 |
| NK $K=8$ | 888 | .231 | .476 | **.099** | 24.0 | 6.0 | **46.1** | 42 | 97 | 16 |
| W $\nu=3$ | 406 | **.037** | .340 | .077 | **90.7** | 23.2 | 77.3 | 15 | 98 | 21 |
| W $\nu=4$ | 3732 | .073 | .258 | **.012** | 73.7 | 20.0 | **97.8** | 13 | 75 | 2 |
| W $\nu=6$ | 5836 | .066 | .221 | **.011** | 68.5 | 18.3 | **96.3** | 14 | 68 | 1 |

these regular Hamming graphs the fitness gradient already captures most of the cross-basin signal. On W-model landscapes, where epistasis distorts the fitness-distance relationship, ORC's advantage over MinGap becomes substantial: 2.31× at $\nu=4$, with only 43% agreement between the two methods.

**When ORC and MinGap disagree, ORC wins.** On W $\nu = 3$, ORC's choice leads to a better basin in **78.7%** of disagreements vs. MinGap's 0.3% (21% ties). On NK $K=4,8$, ORC wins 43–44% vs. MinGap's 24–25%. MinGap almost never outperforms ORC when the two disagree, confirming that ORC's optimal transport matching captures cross-basin structure that the fitness-gap heuristic misses.

**ORC ranks all escape directions.** Spearman correlation between ORC values and destination quality: mean $\rho=0.44$ on W $\nu=3$ (95% positive), $\rho=0.33$ on NK—more negative curvature → better destination basin.

### 5.2 OTG vs. LON: Funnel Detection from Local Data

Both OTG and LON-d1 use only 1-hop data, making this an equal-information comparison. LON-d1 yields 97–98% singleton funnels; OTG compresses to 13–42% funnels with substantially better attractor quality. On W $\nu=3$: OTG rank 0.037 vs. LON-d1's 0.340 (9.2× improvement); 90.7% reach top-5% vs. 23.2%. The OTG even outperforms LON-d3 (3-hop, stochastic) on W $\nu=3$ (rank 0.037 vs. 0.077); at higher epistasis ($\nu \geq 4$), LON-d3's wider perturbation provides superior attractor quality. Path to top-5% attractors: median 2 hops, P90 3 hops.

### 5.3 Scaling Across Problem Sizes and Search Spaces

**MAX-SAT (binary strings).** Proposition 1's sort-and-match reduction lowers per-edge ORC cost from $O(k^3)$ to $O(k \log k)$, enabling computation at 500 optima for $N=100$ in ∼18 s. The ORC/Random escape ratio increases from 2.5× at $N=20$ to 3.7× at $N=100$ ($\alpha=4.27$; $\rho(N, \text{ORC/Rand}) = 0.77$, $p < 10^{-84}$).

**Table 3.** ORC escape advantage on random MAX-SAT. The ORC/Random ratio *increases* with $N$: ORC's distributional signal becomes more valuable in higher dimensions. 20 instances per cell; $\alpha_c \approx 4.27$ is the satisfiability phase transition.

| $N$ | $\alpha$ | %ORC | %Rand | %MG | ORC/R | ORC/MG | Mean ORC |
|---|---|---|---|---|---|---|---|
| 20 | 3.00 | 34.3 | 13.5 | 26.1 | 2.5× | 1.3× | −0.428 |
| 20 | 4.27 | 42.7 | 17.5 | 36.7 | 2.4× | 1.2× | −0.374 |
| 20 | 6.00 | 50.0 | 20.1 | 46.3 | 2.5× | 1.1× | −0.320 |
| 50 | 3.00 | 32.8 | 10.1 | 21.9 | 3.3× | 1.5× | −0.493 |
| 50 | 4.27 | 37.3 | 10.6 | 31.3 | 3.5× | 1.2× | −0.418 |
| 50 | 6.00 | 40.4 | 10.3 | 38.1 | 3.9× | 1.1× | −0.353 |
| 100 | 3.00 | 34.0 | 9.9 | 21.4 | 3.4× | 1.6× | −0.524 |
| 100 | 4.27 | 39.5 | 10.7 | 32.0 | 3.7× | 1.2× | −0.450 |
| 100 | 6.00 | 42.7 | 10.1 | 40.2 | 4.2× | 1.1× | −0.385 |

**Table 4.** OTG vs. LON-d1 on QAP (swap neighborhood, full enumeration). ORC's advantage grows with $n$. LON-d1 degrades to 100% singletons at $n = 9$. 20 instances per row.

| | | | | Escape rate | | Mean rank | | Compression | |
|---|---|---|---|---|---|---|---|---|---|
| $n$ | $k$ | #Opt | ORC/R | ORC | LON | OTG | LON | OTG | LON |
| 6 | 15 | 12 | 1.8× | 51.2 | 4.3 | .230 | .466 | 49% | 96% |
| 7 | 21 | 27 | 2.1× | 55.9 | 3.0 | .235 | .476 | 44% | 97% |
| 8 | 28 | 78 | 2.3× | 58.7 | 0.9 | .230 | .493 | 41% | 99% |
| 9 | 36 | 233 | **2.7**× | 65.0 | 0.1 | **.194** | .499 | 35% | **100%** |

As $N$ grows, the $(N-1)$ exclusive-neighbor comparisons carry more distributional information, strengthening the cross-basin signal. Mean ORC also tracks the SAT phase transition: $\rho(\text{ORC}, \alpha) = 0.94$ ($N = 20$), 0.99 ($N = 100$).

**QAP (permutations).** LON-d1 degrades to 0.1% escape at $n = 9$ (100% singletons, rank $\approx$ random); OTG improves: ORC/Random 1.8× → 2.7×, rank 0.194 with 35% compression. ORC's advantage over MinGap is modest (1.06×) on these random instances; the primary result is OTG vs. LON-d1.

**TSP (2-opt).** Table 5 confirms Corollary 1(iii): the disjoint-neighborhood property holds at all tested sizes. As $k$ grows from 20 to 1175, random escape drops from 25% to 2.5% while ORC maintains 40–45%, yielding an ORC/Random ratio that grows monotonically ($\rho(k, \text{ORC/Rand}) > 0.99$). ORC also consistently outperforms MinGap, winning 45–52% of disagreements vs. 9–22%. This pattern—growing ORC/Random advantage, stable ORC escape rate—is consistent across all three search spaces, confirming that ORC's distributional matching becomes more valuable as neighborhood size increases.

**Table 5.** ORC escape on TSP 2-opt across scales. Top rows: full enumeration. Bottom rows: sampling-based (300 restarts per instance). The ORC/Random ratio grows monotonically from 3.1× at $n=8$ to 18.3× at $n=50$. When ORC and MinGap disagree, ORC wins 45–52% vs. MinGap's 9–22%.

| $n$ | $k$ | #Opt | Escape rate (%) ORC | MG | Rand | ORC/R | ORC/MG |
|---|---|---|---|---|---|---|---|
| *Full enumeration* | | | | | | | |
| 8 | 20 | 4 | 77.5 | 66.7 | 25.0 | 3.1× | 1.16× |
| 9 | 27 | 5 | 64.7 | 57.5 | 21.1 | 3.1× | 1.13× |
| 10 | 35 | 7 | 68.9 | 62.6 | 21.3 | 3.2× | 1.10× |
| *Sampling-based (300 restarts)* | | | | | | | |
| 15 | 90 | 225 | 40.0 | 35.4 | 5.2 | 7.7× | 1.13× |
| 20 | 170 | 289 | 35.7 | 32.9 | 3.9 | 9.2× | 1.09× |
| 30 | 405 | 300 | 40.7 | 38.7 | 3.2 | 12.7× | 1.05× |
| 50 | 1175 | 300 | 45.0 | 43.5 | 2.5 | 18.3× | 1.03× |

**Table 6.** Fixed-budget ILS comparison at $N=16$ (5,000 FEs, 30 instances × 30 trials). ORC+Pert and Rand-ILS both flip 3 bits per perturbation; ORC-ILS and MG-ILS flip only 1 directed bit (ablations). Bold: best per row.

| Config | ORC+Pert | Rand-ILS | RR-HC | ORC-ILS | MG-ILS |
|---|---|---|---|---|---|
| *Success rate (% finding global optimum)* | | | | | |
| NK $K=4$ | 77.3 | **88.9** | 88.4 | 6.7 | 6.1 |
| NK $K=8$ | 42.0 | **44.9** | 45.2 | 1.3 | 1.3 |
| W $\nu=3$ | 46.7 | **80.2** | 52.2 | 8.4 | 1.8 |
| W $\nu=4$ | **89.9** | 69.3 | 39.7 | 1.6 | 1.0 |
| W $\nu=6$ | **87.3** | 56.0 | 26.3 | 1.6 | 0.9 |

### 5.4 ORC-Guided Iterated Local Search

Table 6 compares ILS variants under a fixed 5,000 FE budget. ORC+Pert wins on 2 of 5 configurations and loses on 3, partitioning cleanly along OTG topology. In the *acyclic regime* (NK, W $\nu=3$), Rand-ILS dominates—sometimes by large margins (80.2% vs. 46.7% on W $\nu=3$)—because the landscape has clear funnel structure and ORC's directional constraint reduces exploration diversity. In the *cyclic regime* (W $\nu = 4,6$), ORC+Pert dominates: 89.9% vs. 69.3% at $\nu = 4$. ORC-ILS and MG-ILS, which flip only a single directed bit, perform poorly (1–8% success) because without random bits their perturbation deterministically revisits the same path and becomes trapped in OTG cycles.

**Computational cost.** At $N = 16$, ORC costs ∼120 FEs (∼3 ms); at $N = 100$, ∼5 000 FEs (∼18 ms). At $N = 100$ on MAX-SAT (10 s wall-clock budget), Rand-ILS completes 25× more iterations and finds better solutions; ORC's 3.7× per-step advantage cannot overcome the iteration deficit when evaluations cost

**Table 7.** Spearman correlations ($\rho$) between landscape metrics and algorithm performance ($N = 16$, 480 instances: 30 each for NK $K \in \{4, 8, 12, 15\}$ and W-model $\nu \in \{1, \ldots, 12\}$). $^{***}$: $p < 0.001$. Best per column in **bold**. †: requires full enumeration. HC: best-improvement hill climbing (success = global optimum found). EA: ($\mu$+$\lambda$) evolutionary algorithm with $\mu$=5, $\lambda$=10, bit-flip mutation (success rate). RS: random sampling baseline (mean fitness of $10^4$ uniform samples).

| Feature | Cost | HC succ. | EA succ. | RS mean |
|---|---|---|---|---|
| Mean ORC (ours) | $O(Rk^2)$ | $-0.670^{***}$ | $\mathbf{-0.652}^{***}$ | $-0.455^{***}$ |
| %ORC→better | $O(Rk^2)$ | $-0.435^{***}$ | $-0.174^{***}$ | $-0.615^{***}$ |
| FDC | $O(2^N)$ | $0.276^{***}$ | $0.501^{***}$ | 0.024 |
| Autocorrelation | $O(L)$ | $0.543^{***}$ | $0.544^{***}$ | $0.372^{***}$ |
| Info. content $H$ | $O(L)$ | $-0.308^{***}$ | $0.306^{***}$ | $-0.585^{***}$ |
| Mean $\lvert\Delta f\rvert$ (ELA) | $O(L)$ | $\mathbf{-0.695}^{***}$ | $-0.083$ | $\mathbf{-0.769}^{***}$ |
| NBC mean (ELA) | $O(L)$ | $-0.419^{***}$ | 0.072 | $-0.487^{***}$ |
| Basin entropy† | $O(2^N)$ | $-0.731^{***}$ | $-0.460^{***}$ | $-0.610^{***}$ |
| OTG attractor rank† | $O(2^N)$ | $0.636^{***}$ | 0.157 | $0.754^{***}$ |
| OTG compression† | $O(2^N)$ | $0.589^{***}$ | $-0.094$ | $0.748^{***}$ |

$\sim$0.001 ms. ORC-guided search becomes cost-effective when individual evaluations cost $\geq 1$ ms (e.g., simulation-based optimization).

## 6 Discussion

### 6.1 OTG Structural Properties

Figure 1 reveals a structural crossover: at W $\nu = 3$ the OTG is acyclic (91% top-5%); at $\nu$=4, cycles emerge trapping 21% of optima (top-5% drops to 74%). NK landscapes show no cycles at any $K$. This crossover correlates with the ILS results in Table 6: in acyclic regimes, random perturbation suffices and ORC's directional constraint reduces diversity; in cyclic regimes, ORC+Pert's directed bit breaks cycles that trap random perturbation. We note this is an observed correlation on the tested benchmark families; validating it as a predictive criterion on unseen landscape types is an avenue for future work.

### 6.2 ORC as Difficulty Predictor

Table 7 evaluates ORC as a difficulty predictor alongside classical FLA metrics. Mean $|\Delta f|$ achieves the strongest individual correlation on HC ($\rho = -0.695$) and RS ($\rho = -0.769$), but fails on EA ($\rho = -0.083$, not significant). Information content and NBC show similar target-specificity, with sign reversals across algorithm types. Autocorrelation maintains consistent sign but with weaker magnitude on RS ($\rho = 0.372$). Mean ORC is the only tractable metric with strong correlations (all $|\rho| > 0.45$) of consistent sign across all three algorithm types, making it a

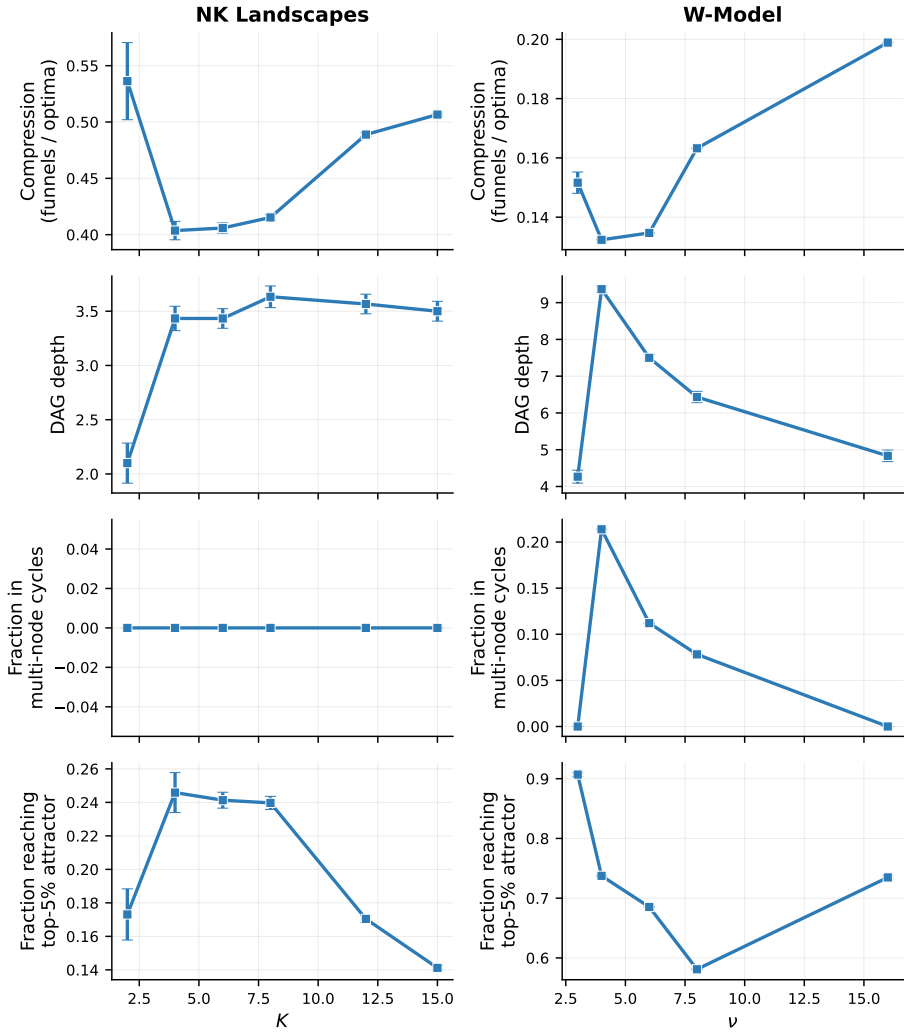


**Fig. 1.** OTG structural properties vs. ruggedness parameter. Left column: NK landscapes ($K$). Right column: W-model ($\nu$). A structural crossover in cycle structure appears at W $\nu = 4$: multi-node cycles emerge (21% of optima), DAG depth peaks, and top-5% reachability drops. NK landscapes show no cycles at any $K$.

robust general-purpose predictor that does not require choosing the right metric for a specific algorithm family.

Table 8 extends the escape analysis to $N = 20$, confirming the pattern beyond the $N = 16$ regime: ORC's advantage decreases monotonically with $K/N$ as maximal-ruggedness landscapes leave no exploitable distributional structure.

**Table 8.** ORC escape at $N=20$ (NK landscapes, full enumeration, 10 instances each). The ORC/Random advantage replicates the $N = 16$ pattern; at $K = 19$ ($\approx N$), all neighbors are equally rugged and ORC's advantage vanishes.

| $K$ | #Opt | %ORC | %Rand | ORC/R |
|---|---|---|---|---|
| 4 | 557 | 62.3 | 31.2 | 2.0× |
| 8 | 5,012 | 62.5 | 34.1 | 1.8× |
| 15 | 26,086 | 52.9 | 37.2 | 1.4× |
| 19 | 49,966 | 49.5 | 47.5 | 1.0× |

### 6.3 OTG Visualization

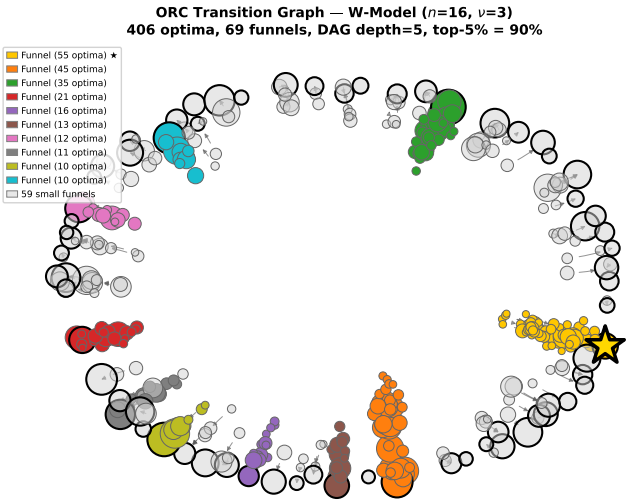


**Fig. 2.** OTG for a W-model instance ($N = 16$, $\nu = 3$, seed 0). Each node is a local optimum, sized by basin of attraction and colored by funnel membership. The gold star marks the global optimum. Edges follow the min-ORC direction deterministically. The 406 optima collapse into 69 funnels; 91% converge to a top-5% attractor. The circular layout groups funnels; spatial distance does not encode fitness or Hamming distance.

Figure 2 visualizes the OTG for a representative W-model instance. Clusters of optima flow deterministically toward high-quality attractors from local curvature alone, without stochastic trials or perturbation distance choices.

### 6.4 Convergence Under Sampling

On W-model at $N=16$, subsampling $R$ optima uniformly (20 subsamples $\times$ 10 instances), per-optimum metrics converge fast: escape rate within 2% error at $R=5$, <1% at $R=20$. Structural metrics converge slowly—at 50% coverage, attractor rank is still 3.4$\times$ worse than ground truth, because the OTG cannot route to unsampled attractors. Per-optimum ORC metrics are thus reliable from $\sim$20 restarts at any scale; full OTG structural analysis remains limited to enumerable landscapes.

### 6.5 Limitations

The strongest OTG results (up to 9.2$\times$ attractor quality improvement over LON-d1) come from full enumeration at $N=16$ on W-model—a scale where exhaustive analysis is feasible but not representative of real-world problem sizes. At $N=100$ on MAX-SAT, Rand-ILS outperforms ORC+Pert in wall-clock time when evaluations are cheap ($\ll$1 ms). On QAP, ORC's advantage over MinGap is marginal (1.06$\times$) on random instances; structured QAPLIB instances may behave differently. For neighborhoods *without* the disjoint-neighborhood property (e.g., 3-opt, LK moves), Proposition 1's decomposition does not apply and ORC requires the full $O(k^3)$ Hungarian algorithm; characterizing ORC's behavior on such graphs is an open question. A full comparison with the flacco/ELA ecosystem [8] in an algorithm selection pipeline remains future work.

## 7 Conclusion

The ORC Transition Graph demonstrates that optimal transport over neighborhood fitness distributions captures cross-basin structure invisible to 1-hop methods, with the attractor quality gap over LON-d1 reaching up to 9.2$\times$ on epistatic W-model landscapes at $N = 16$. The disjoint-neighborhood property (Corollary 1) gives the framework a unified foundation across the three foundational neighborhoods in combinatorial optimization: bit-flips, permutation swaps, and TSP 2-opt. ORC's per-step escape advantage grows monotonically with neighborhood size (3.1$\times$ to 18.3$\times$ over random on TSP from $n=8$ to $n=50$; analogous trends on binary and swap spaces), though translating this into algorithmic gains remains cost-dependent. OTG structural properties—particularly the acyclic-to-cyclic crossover—correlate with when ORC-guided perturbation adds value in ILS, providing practitioners with a diagnostic signal. Per-optimum metrics converge from $\sim$20 samples, making ORC a practical addition to the FLA toolkit.

*Reproducibility.* All code and data are available at `https://github.com/RobertoWTBW/RiemannianSwarmSurgery`.

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